



MEDICAPS
U N I V E R S I T Y

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GEARGENIX: Advanced Campus Gear Control

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE
& ENGINEERING

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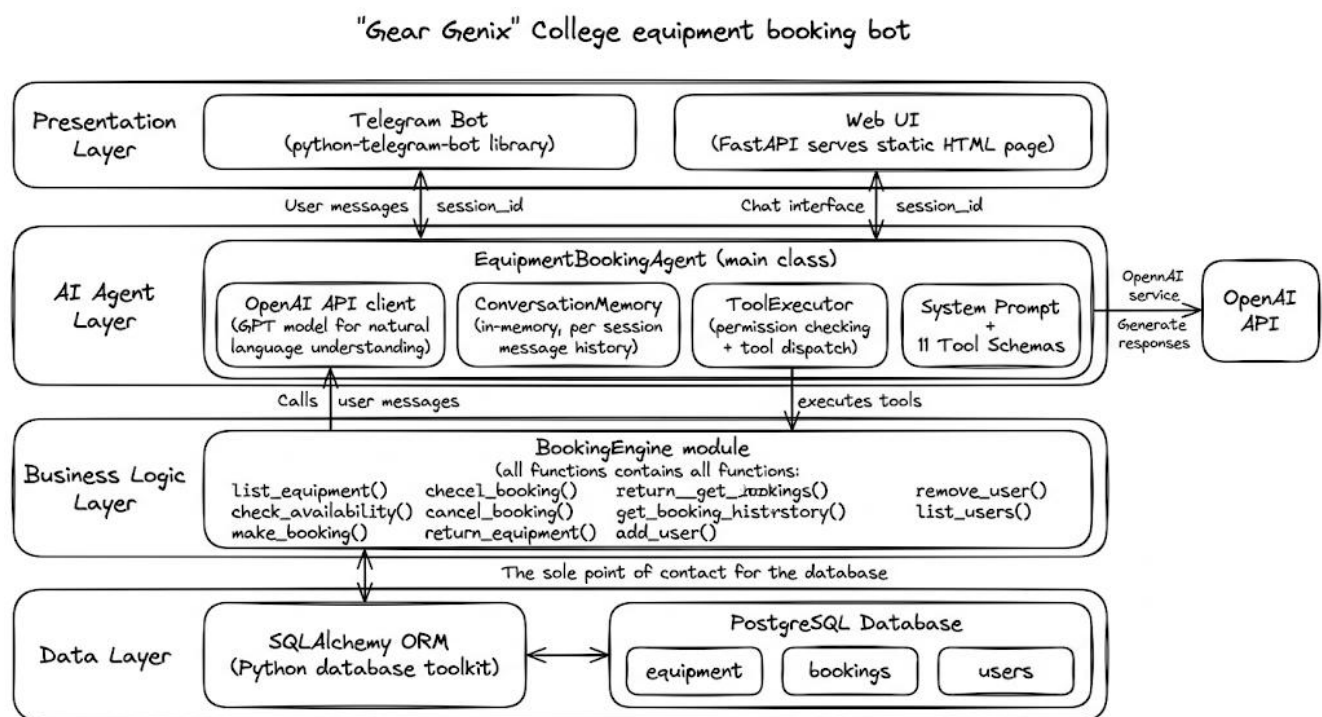
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1. Introduction

GearGenix is an AI-based equipment booking system designed for colleges to manage shared resources like projectors, microphones, laptops, and speakers. In most colleges, booking is done through WhatsApp messages or manual registers, which leads to confusion, double bookings, and no proper tracking. GearGenix solves this problem by providing a simple system where users can book equipment using natural language, just like sending a message.

The system uses an AI agent to understand user requests, check availability in real time, and complete the booking process automatically. It can be accessed through a Telegram bot or a web interface, making it easy for both students and admins to use. The backend is built using FastAPI, data is stored in PostgreSQL, and the web interface is created using React. Overall, the system reduces manual work, improves transparency, and ensures smooth equipment management.

2. System Architecture

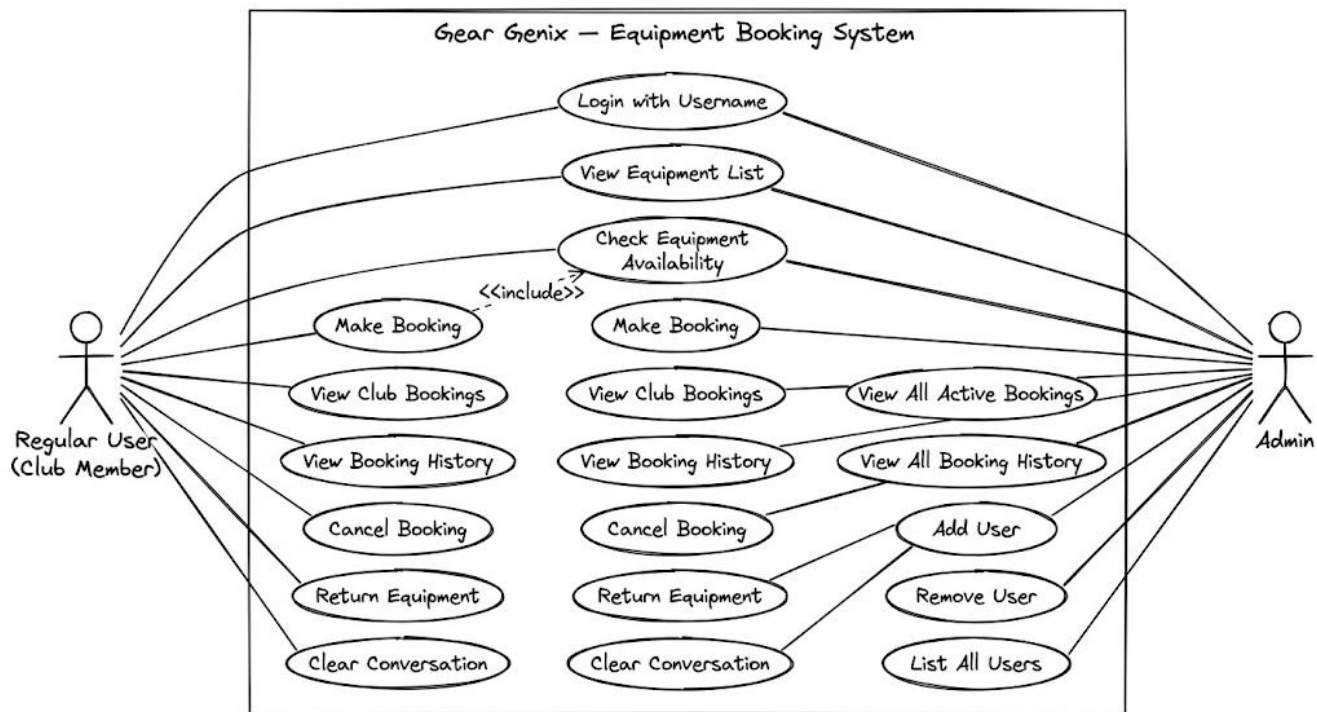


The system follows a three-layer architecture consisting of the presentation layer, application layer, and data layer. The presentation layer includes the Telegram bot and the web interface, which act as entry points for users to interact with the system. These interfaces send user requests to the backend.

The application layer is handled by a FastAPI server, which processes all requests and manages the main logic of the system. It also connects with the AI module, which understands the user's message and extracts important details like equipment name and time. The data layer consists of a PostgreSQL database that stores all information related to users, equipment, and bookings.

The flow is simple: the user sends a request, the backend processes it with the help of AI, the database is checked or updated, and a response is sent back to the user.

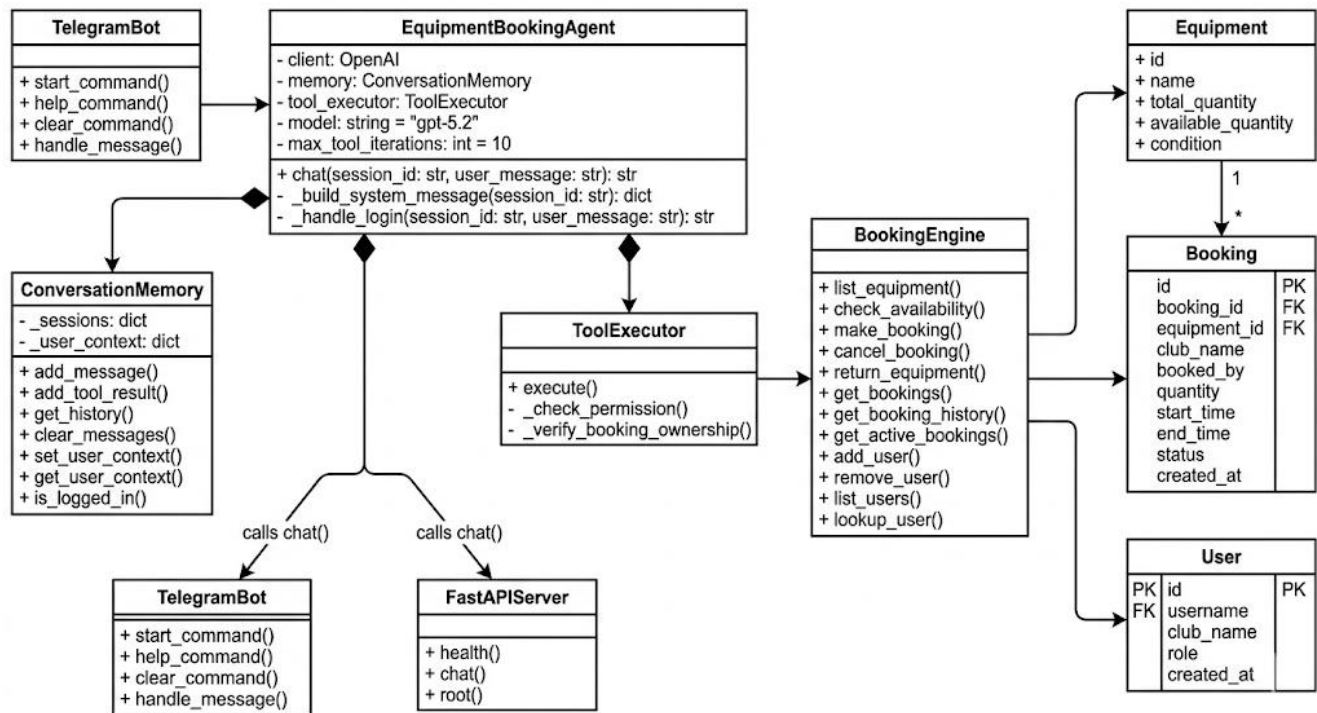
3. Use Case Diagram



The use case diagram shows how different users interact with the system. There are two main actors: the student (or user) and the admin. Students use the system to perform actions like viewing available equipment, checking availability, booking items, cancelling bookings, and returning equipment after use.

The admin has additional responsibilities such as viewing all bookings and managing equipment records. One important point is that booking equipment always includes checking availability first, so the system ensures that no conflicts occur. This diagram helps in understanding all the main functions provided by GearGenix and how users interact with them.

4. Class Diagram

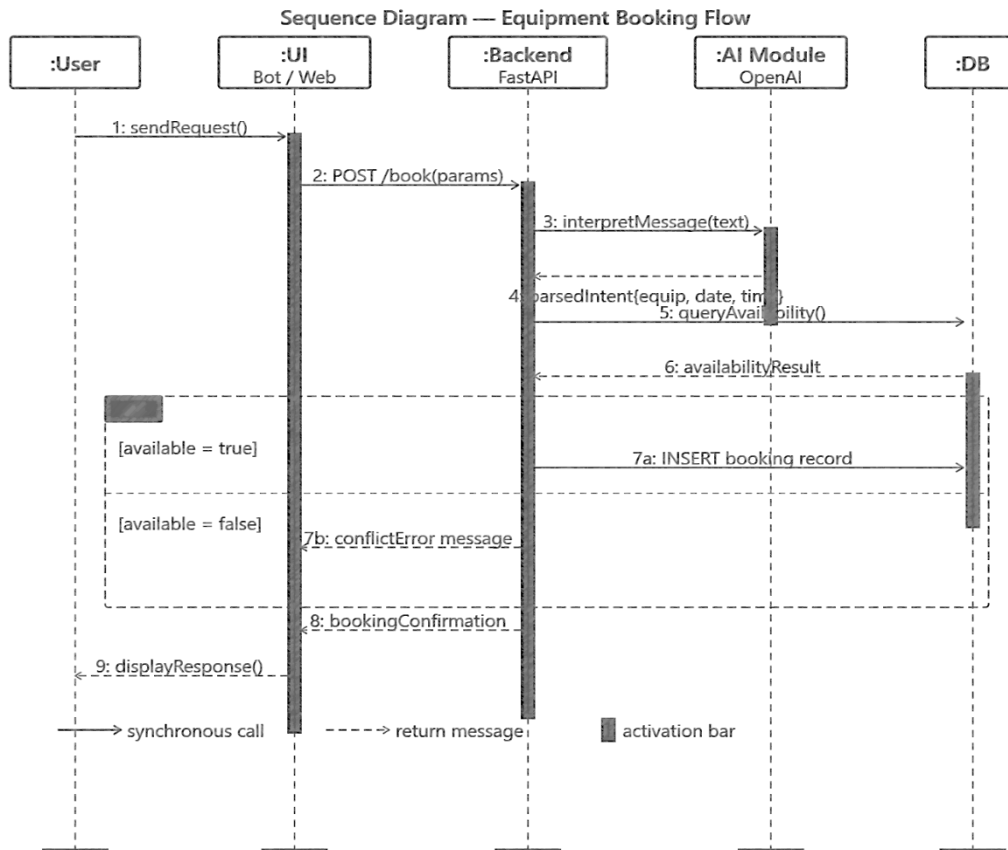


The class diagram shows the main structure of the GearGenix system with key classes: User, Admin, Booking, Equipment, and ConflictChecker.

The User class handles actions like login, booking, and returning equipment, while the Admin class extends User and adds functions like managing equipment and viewing all bookings. The Booking class connects users and equipment and stores details like date, time, and status. The Equipment class stores information about items and checks their availability. The ConflictChecker ensures that no two bookings overlap.

Overall, the diagram shows how different parts of the system are connected and work together to manage equipment booking efficiently.

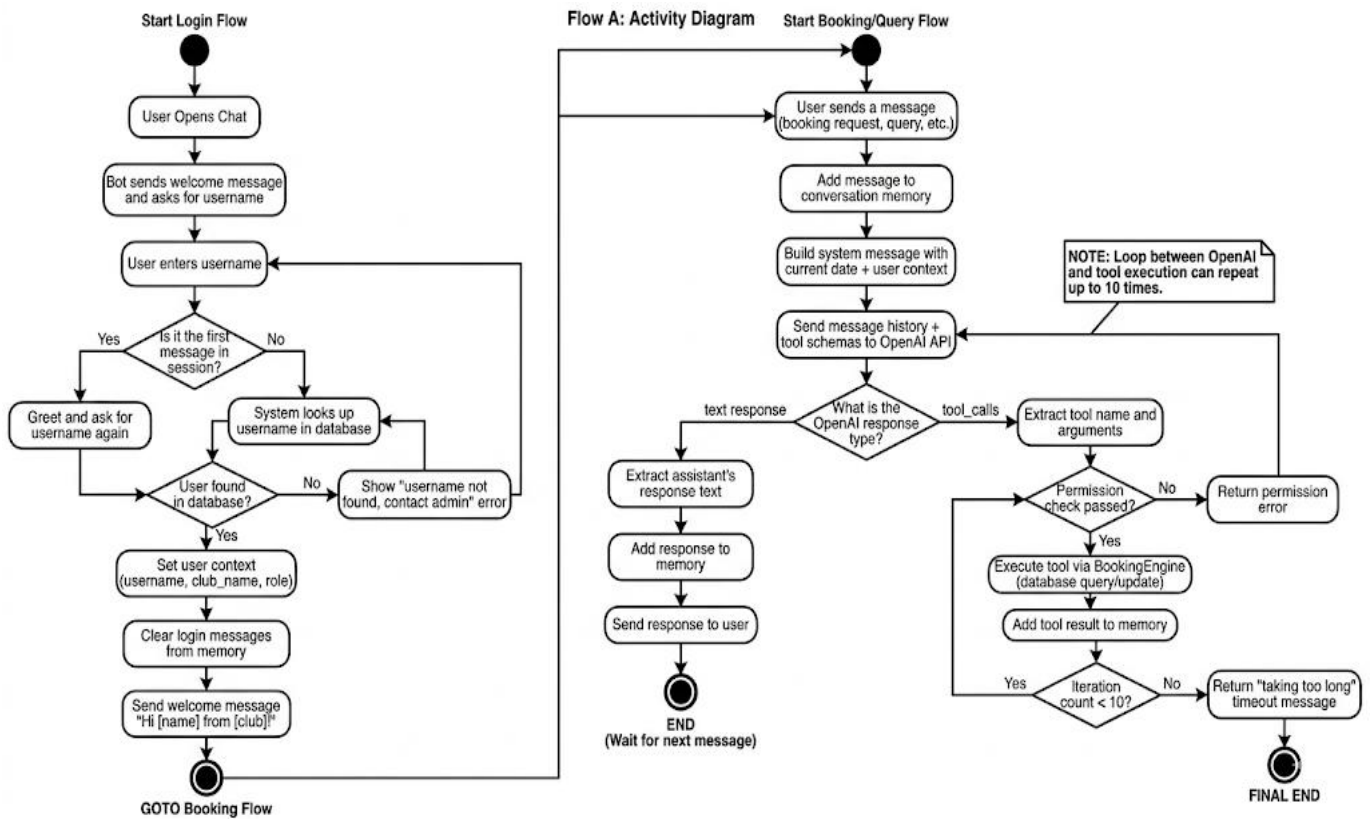
5. Sequence Diagram



The sequence diagram explains the step-by-step process of booking equipment. First, the user sends a request through the Telegram bot or web interface. This request is forwarded to the backend server, which then sends it to the AI module to understand the user's intention.

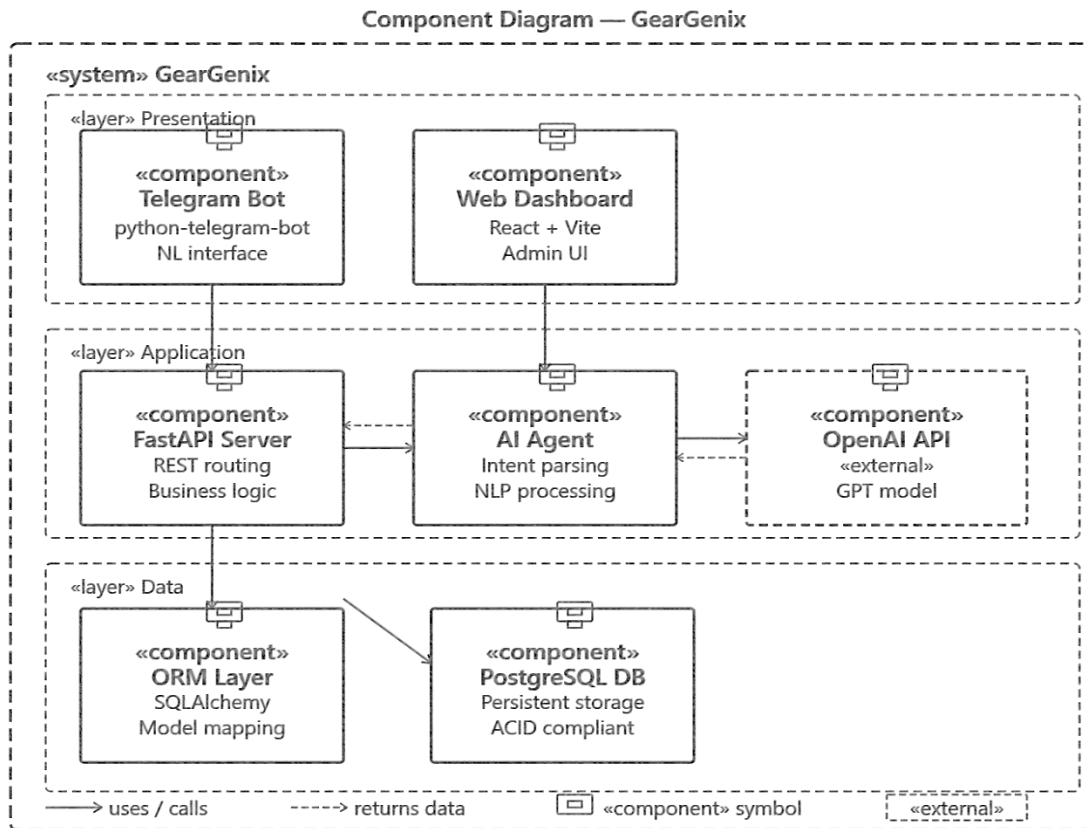
After extracting the required details, the backend checks the database to see if the equipment is available for the requested time. If the item is available, a new booking is created and stored in the database. If it is not available, the system prepares a conflict message. Finally, the result is sent back to the user through the interface. This diagram shows how different components communicate in order to complete a booking.

6. Activity Diagram



The activity diagram shows the complete workflow of booking equipment in the GearGenix system. The process starts when the user opens the bot or web app and sends a booking request. The UI receives the request and forwards it to the backend, where the AI processes the message and extracts details. The system then checks the database for equipment availability. If the item is available, a booking record is created; otherwise, a rejection message is generated. Finally, the response is sent back to the user and displayed on the interface. This diagram clearly shows the step-by-step flow and decision-making involved in the booking process.

7. Component Diagram



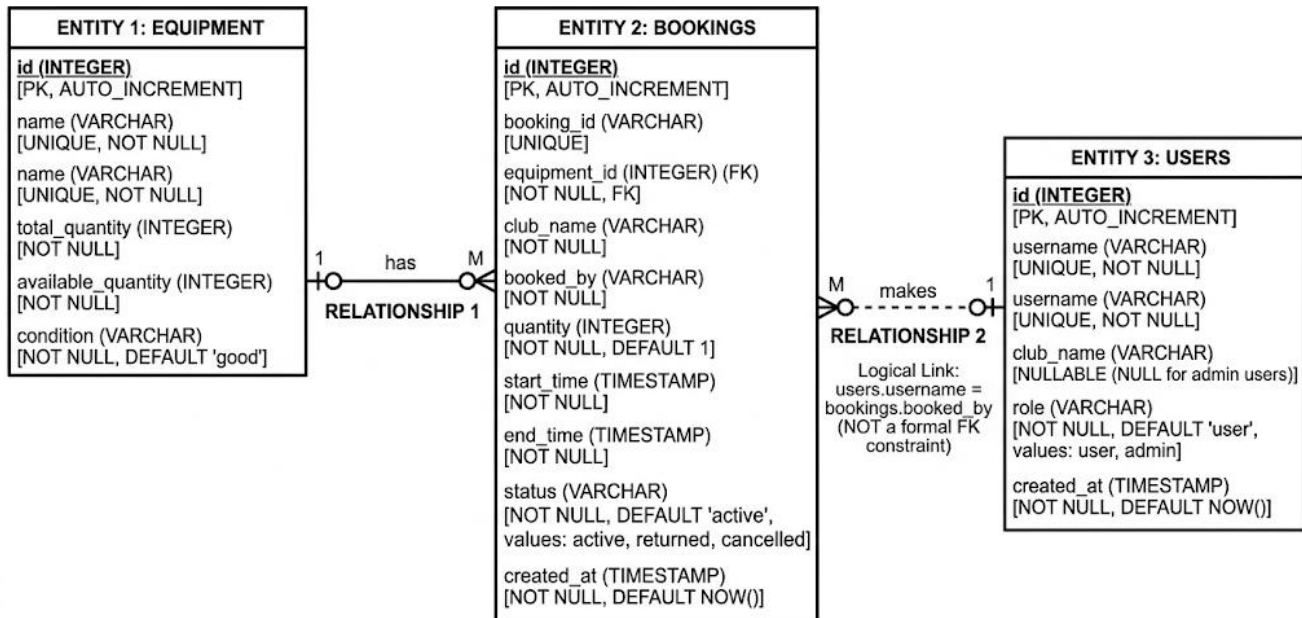
The component diagram shows the major parts of the GearGenix system and how they are connected. It is divided into three layers: presentation, application, and data.

The presentation layer includes the Telegram bot and the web dashboard, which allow users to interact with the system. These components send user requests to the backend. The application layer contains the FastAPI server, which handles all business logic, and the AI agent, which processes user messages and extracts useful information. The system also connects to the OpenAI API for natural language understanding.

The data layer includes the ORM layer and the PostgreSQL database, which store and manage all system data such as users, equipment, and bookings. Overall, this diagram shows how different components work together to process requests and provide responses efficiently.

8. ER Diagram

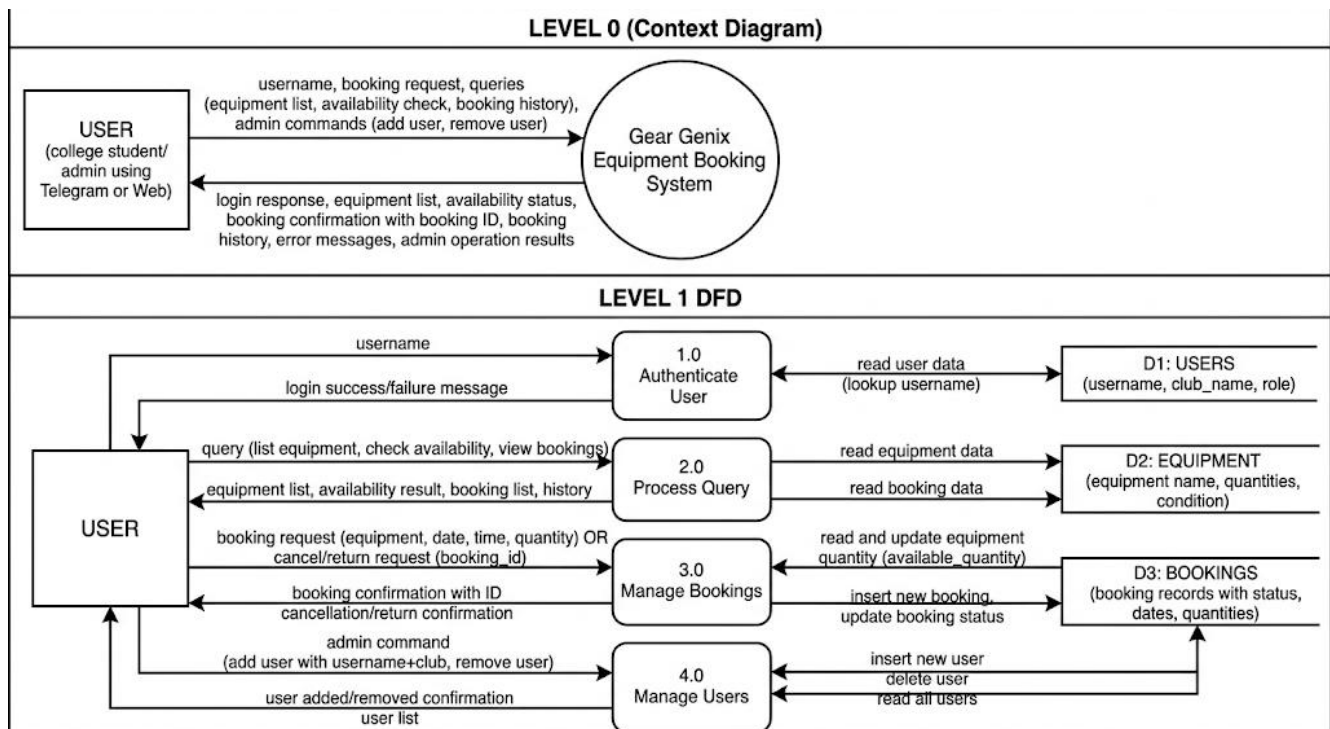
GEAR GENIX EQUIPMENT BOOKING SYSTEM ER DIAGRAM



The ER diagram represents how data is structured in the system. There are three main entities: User, Equipment, and Booking. The User entity stores details of students or admins. The Equipment entity stores information about all available items. The Booking entity records each reservation made in the system.

A user can make multiple bookings, which shows a one-to-many relationship between User and Booking. Each booking is linked to one piece of equipment, showing a many-to-one relationship between Booking and Equipment. This design helps in organizing data clearly and ensures that all bookings are properly tracked and managed.

9. DFD Diagram



The data flow diagram shows how information moves through the GearGenix system. At the top level, the system interacts with two external entities: the user (student) and the admin. The user sends booking requests and receives confirmation or status updates, while the admin manages equipment and views bookings.

At the next level, the system is divided into main processes such as user authentication, request processing, availability checking, booking creation, and sending notifications. When a user sends a request, it is first authenticated, then processed by the system, and checked for availability. If there is no conflict, the booking is stored in the database; otherwise, a conflict is detected.

The system uses data stores like user data, equipment data, and booking records to read and write information. Finally, the processed result is sent back to the user as a confirmation or rejection message. This diagram clearly shows how data flows between different parts of the system.